

DTSC 5302

Ethical Issues in Data Science

FORMAT	CATEGORY	CREDITS
On-Campus	Core	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 2 – Vital Skills (on-campus)	Machine Learning Engineer (43.3 / 100)	36.6 / 100
SPECIALIZATION General Data Science (on-campus)		

37

CURRENT AVG SCORE
across 6 job roles

70

POTENTIAL SCORE
with recommended additions

+44

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

This course examines ethical issues related to data science, with the objective of making data science professionals aware of and sensitive to ethical considerations that may arise in their careers. It focuses on ethical frameworks, data science applications that lead to ethical considerations, current media and scholarly articles, and the perspectives and experiences of fellow students and computing professionals.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	43.3	68.4	+25.1	MEDIUM
AI Engineer	38.9	75.1	+36.2	HIGH
Data Scientist	36.8	61.4	+24.6	MEDIUM
ML Engineer	36.4	66.4	+30.0	HIGH
MLOps Engineer	35.6	77.9	+42.3	HIGH
Applied Scientist	28.8	72.8	+43.9	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design

5 of 6 roles

systems

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1

HIGH

Add System Design coverage — required vocabulary in 5 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

GAP KEYWORDS BY ROLE

Machine Learning Engineer	43.3 → 68.4 (+25.1 pts)
learning tools design models model machine technical systems	
AI Engineer	38.9 → 75.1 (+36.2 pts)
robotics models learning technical model systems teams problems	
Data Scientist	36.8 → 61.4 (+24.6 pts)
social people problems	
ML Engineer	36.4 → 66.4 (+30.0 pts)
learning machine systems models model world problems impact	
MLOps Engineer	35.6 → 77.9 (+42.3 pts)
detection model scientists systems learning machine image tools	
Applied Scientist	28.8 → 72.8 (+43.9 pts)
computational research problems models technical learning systems industry	

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.